

Displacement-enforced mode transitions in rotary metamorphic flexure bearings

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Abstract

Bearings play a critical role in mechanical and mechatronic systems. Flexure bearings enable high-precision rotational motion, and their wear-free properties have the potential to enable substantially more longevity than conventional bearings. However, flexure bearings are typically only able to operate over a limited range of motion, precluding their use in applications requiring larger displacements. To address this, we previously introduced the metamorphic flexure bearing (MFB), which primarily operates as a flexure bearing but transitions automatically to a conventional bearing to accommodate greater displacement. The transition between bearing modes is enforced mechanically by a retention mechanism, which produces a displacement-dependent resistive force to selectively activate one bearing at a time. This dual-mode design imbues the MFB with the high precision and low-wear properties expected of flexure bearings, with the conventional bearing's extended range of motion. Herein, we present a rotary-motion MFB and demonstrate key performance characteristics during quasi-static operation for two types of retention mechanisms (retention-handoff and constant-stiffness). We also investigate the occurrence of a “combined mode”, in which both bearings are simultaneously engaged during dynamic operation, and define the acceleration thresholds that govern this behavior. Our results provide a starting point for design and implementation of rotary MFBs across applications.

Keywords

complaint mechanisms, stiffness modeling, transient dynamic analysis, experimental verification

1. Introduction

Bearings are critical mechanical components that guide motion in mechanical systems [1], [2]. Conventional bearings such as ball, thrust, roller, plain, and journal bearings all facilitate infinite or near-infinite angular displacements, but introduce friction or rolling contact at articulating surfaces [3]. State-of-the-art conventional mechanical bearings have improved substantially over the past century [4], [5], but the inevitable friction or rolling contact of a conventional bearing reduces the system's efficiency and limits lifespan [6], eventually necessitating maintenance and repair [7] when considering long-term use. Although lubrication can mitigate the adverse effects of friction [8], lubricants may be unsuitable for environments that are sensitive to particulates, such as in a clean room or in an implantable medical device. Bearing failure is particularly challenging in high-load environments and situations with limited maintenance access where repairs are difficult, such as inside of complex machines [9], deep underwater [10], in cryogenic environments [11], or in space [12], in which bearing longevity and precision are high priorities.

As an alternative to conventional bearings, flexure bearings guide motion through elastic deformation of thin blades [13], [14], thereby eliminating friction- and contact-related wear and inefficiencies. While this reliance on deformation necessitates careful fatigue considerations, a flexure designed such that its maximum operating stress

remains below the material's endurance limit can have increased longevity [15], [16]. Flexure bearings can also guide motion with extremely high precision and are thus well-suited for fine-resolution systems such as nano positioning [17], optical alignment [18], and micro-electromechanical sensors [19]. A key limitation of flexure bearings is that their range of motion is inherently constrained [20], such that they can only be used in applications that require relatively small displacements. If it were possible to extend this range of motion, the advantages of flexure bearings could potentially be applied to increase the operational lifespan and precision of machines that require large displacement.

As a step toward this ideal, we previously introduced the metamorphic flexure bearing [21], which functions as a flexure bearing over its primary range of motion and automatically transitions to a conventional bearing when extended range is needed. Transition between the two bearing modes is governed by a retention mechanism, which ensures that only one of the two bearing modes is active at any given time. A comparison of the expected performance of metamorphic flexure bearings to conventional and flexure bearings is provided in Table 1. In our prior work, we introduced three categories of retention mechanisms and provided examples of each: constant stiffness retention (compression spring), constant force retention (constant-force spring), and retention handoff (magnetic catches). We then applied these in a prototype metamorphic bearing for *linear* motion and characterized the quasi-static behaviors of each retention mechanism type. However, many applications require rotary motion, so extension of the metamorphic flexure bearing to a rotary design has the potential to provide these advantages to high-cycle-count cyclic rotary motion applications.

	Conventional Bearings	Flexure Bearings	Metamorphic Flexure Bearings
Range of Motion	High	Low	Moderate
Wear	Moderate-High	Low	Low-Moderate
Precision	Moderate	High	Moderate-High
Complexity	Low	Moderate	High
Volume	Low	Moderate	High
Dynamics	Friction	Stiffness	Stiffness/Transition/Friction
Durability	High	High*	High*

Table 1. Comparison of expected performance characteristics of metamorphic flexure bearings with conventional and flexure bearings. We list here the expected advantages and disadvantages of metamorphic flexure bearings relative to conventional bearings and flexure bearings. Note that the ratings assigned here are generalized, whereas true specifications of a bearing are highly design-specific and application-specific. For instance, a linear metamorphic flexure bearing (MFB) can have an infinite range of motion, while a rotary MFB cannot, the wear on an MFB is dependent on how much time it spends in its extended range, and the durability of a flexure bearing is highly dependent on proper hard-stop synthesis *[22].

In this work, we extend the principles of the metamorphic flexure bearing to *rotational* bearings, presenting the design and validation of a prototype rotary metamorphic flexure bearing (rotary MFB). This extension requires redesigning the bearing architecture to support series transition between the rotational flexure and conventional bearings, as well as integration of retention mechanisms to mediate mode switching. Additionally, our previous work [21] did not test dynamic scenarios. In this work, we address both of these challenges. As part of this process, we redesigned two of the retention mechanisms from our previous work [21] (“retention handoff” and “constant stiffness retention”) for rotational motion and evaluated their performance in the rotary MFB prototype. In addition, we had previously postulated that rapid acceleration of the mechanism could induce a mechanically ambiguous

“combined mode” in which both bearings are active simultaneously. In this manuscript, we provide experimental evidence of this combined mode and seek to understand how it affects the bearing’s behavior.

2. Methods

2.1. Principles of Operation

We will first provide a brief introduction to the principles of metamorphic flexure bearing design and describe how they apply to a rotational system; for a full theoretical treatment of these principles, we refer the reader to our previous work [21]. A metamorphic flexure bearing consists of three rigid bodies: a stage body, an intermediate body, and a ground body. A flexure bearing (teal, Fig. 1A) connects the stage body to the intermediate body, which is then connected to the ground body through a conventional bearing (yellow, Fig. 1A). The metamorphic bearing operates under two primary working modes: flexure-bearing mode and conventional-bearing mode. While in flexure-bearing mode, motion of the stage body is guided by a compliant mechanism, and the intermediate and ground bodies remain rigidly coupled. When the stage body is moved beyond the flexure-bearing’s range of motion, the device automatically transitions into conventional-bearing mode. In conventional-bearing mode, the stage and intermediate body are rigidly coupled, and the conventional bearing guides movement of these coupled bodies relative to the ground body. Internal hard stops enforce the angular range of motion for each bearing, and retention mechanisms linked to the hard stops enforce coupling of the different bodies.

Metamorphic flexure bearings are best suited for applications that are centered in a particular range of motion but have an occasional need for larger displacement. When compared to conventional bearings in these applications, the rotary MFB is expected to reduce contact-related wear when operating within the operating range of the flexure bearing, potentially extending overall bearing longevity when the flexure bearing is optimized for high cycle counts. Traditional flexure bearings can provide this benefit at the expense of range of motion, but the rotary MFB could provide this benefit along with access to a larger range as needed. Our primary design objectives in this work were to provide a sufficient angular range for testing throughout which the *flexure* bearing was active and to ensure smooth and reliable transition between modes. Both properties depend heavily on the retention mechanism. From the handful of mechanisms that can apply retention force from our previous work [21], we focus herein on “retention handoff” and “constant stiffness retention”.

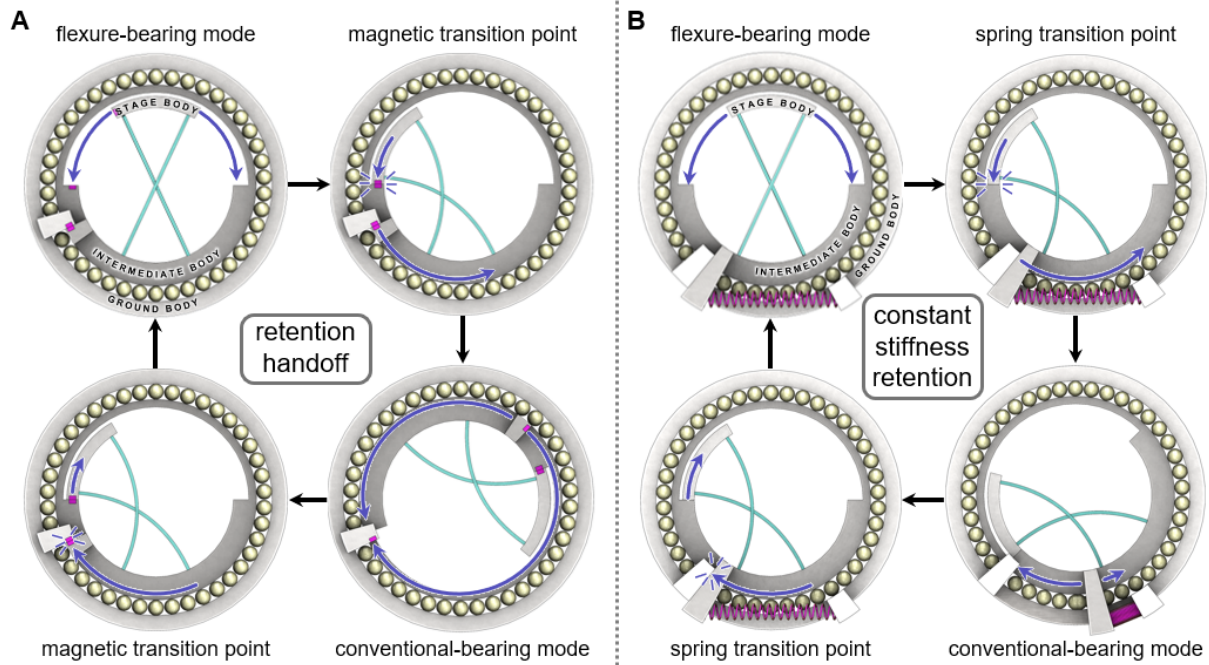


Fig. 1. Rotary metamorphic flexure bearing design principles. The rotary MFB consists of at least three rigid bodies - the stage body, the intermediate body, and the ground body. A flexure bearing, highlighted in teal, connects the stage body and intermediate body. A conventional bearing, highlighted in yellow, connects the intermediate body and ground body. Transition mechanisms are shown engaging in a secondary layer via elements that are rigidly connected to elements in the first layer. Highlighted in magenta are two transition mechanism types: (A) retention handoff using magnetic pairs and (B) constant stiffness (linear torque) retention using a spring (here shown diagrammatically as a compression coil spring, though in this work we use a torsion spring for greater range of motion).

Retention handoff mechanisms enforce rigid body coupling by “passing” retention of the intermediate body between the stage body and the ground body. One advantage of retention handoff is that it introduces only transient changes to the force profiles of the bearings, at the point of handoff. In this paper we use pairs of permanent magnets (magenta, Fig. 1A) to implement retention handoff. The stage pair is located between the stage body and intermediate body, and the ground pair is located between the intermediate body and ground body. In flexure-bearing mode, which serves as the primary operational mode (top-left configuration, Fig. 1A), the stage-pair magnets are initially disengaged, and the ground-pair magnets are initially engaged and coupling the intermediate and ground bodies. Throughout the flexure bearing’s angular range of motion, the rotary MFB behaves purely as a compliant mechanism. As the bearing passes through the magnetic transition point, the stage-pair magnets engage and the ground-pair magnets disengage, creating a disturbance in the bearing’s force-displacement profile. After the transition, the rotary MFB acts purely as a conventional bearing. In this way, the force-displacement effects of the magnetic pairs are isolated to within the transition zone, preserving the discrete flexure- and conventional-bearing-mode behaviors outside of this region. When the joint returns to the flexure bearing’s angular range of motion, the handoff occurs in the opposite direction, such that the rotary MFB reverts automatically to flexure-bearing mode.

Constant stiffness retention mechanisms enforce rigid body coupling through a sustained resistive force or torque that increases linearly with displacement of the conventional bearing. In this paper, we use a preloaded rotational spring to enforce constant stiffness retention (magenta, Fig. 1B). Throughout the flexure bearing’s angular range of motion, the rotary MFB behaves purely as a compliant mechanism, with the resistive torque from the preloaded rotational spring enforcing a rigid coupling between intermediate and ground bodies. As the rotary MFB passes through the spring transition point, the torque applied on the spring exceeds the resistive preload on the spring,

allowing the intermediate body to decouple from the ground body. Simultaneously, the contact with the hard stop at the end of the flexure bearing range of motion couples the stage body to the intermediate body. Following this transition, the rotary MFB acts as a parallel combination of the conventional bearing and the torsion spring. In contrast to the retention handoff mechanism, which imparts a localized force only within the transition zone, the constant stiffness retention mechanism imposes a continuous angle-dependent torque throughout the compliant bearing's full range of motion.

A key principle across all retention mechanisms for rotary MFBs is that the retention torque must exceed the torque required to deform the flexure bearing to the end of its range of motion. This is a minimum threshold requirement that, in quasi-static operation, ensures rigid coupling between the intermediate and ground bodies during flexure bearing mode, and between the stage and intermediate bodies during conventional-bearing mode. The retention torque required to ensure single-mode operation increases with the dynamic performance requirements of the rotary MFB, due to the non-zero inertia of the intermediate body. If the retention torque is insufficient to maintain rigid coupling at any instant in the trajectory, the device can temporarily enter a combined mode in which the intermediate body is not rigidly coupled to either the stage or ground bodies.

The dynamic equilibrium of the system is defined by the following relation:

$$\tau_r = J\ddot{\theta} + \tau_f(\theta) + \tau_d \quad (1)$$

Where τ_r is the retention torque provided by the retention mechanism, J is the rotational inertia of the moving stage, $\ddot{\theta}$ is the angular acceleration, τ_f is the flexure restoring torque, and τ_d are external torque disturbances.

To establish a normalized metric for design optimization, we defined two distinct dimensionless ratios. The inertial-to-retention ratio, R_1 , represents the portion of the retention torque required to overcome the system's inertia during maximum acceleration:

$$R_1 = \frac{J\ddot{\theta}_{max}}{\tau_r} \quad (2)$$

The flexure-to-retention torque ratio, R_2 , represents the portion of the retention torque required to overcome the structural stiffness of the flexure:

$$R_2 = \frac{\tau_f(\theta)}{\tau_r} \quad (3)$$

The remaining margin between these ratios is the normalized external torque disturbance tolerance, Π_d :

$$\Pi_d = \frac{\tau_d}{\tau_r} = 1 - R_1 - R_2 \quad (4)$$

Evaluating the boundary conditions of this metric reveals two critical insights. First, at the critical threshold, $\Pi_d = 0$, the retention torque is exactly enough to overcome the internal inertial and flexure torques. Consequently, the system has zero extra tolerance for external torque, and any such disturbance would force the rotary MFB into a combined mode. Second, at the maximum robustness, $\Pi_d \rightarrow 1$, the internal inertial and flexure torques are negligible compared to the retention torque, resulting in a highly robust system that can withstand high external disturbances.

Designing a rotary MFB requires estimating both the peak angular acceleration and maximum anticipated disturbance torque in the system. The retention torque must be engineered to satisfy the following relation:

$$\tau_r \geq J\ddot{\theta}_{max} + \tau_f(\theta) + \tau_d \quad (5)$$

2.2. Rotary MFB Design

When designing metamorphic flexure bearings for specific applications, the bearing architecture can be tailored to meet task-specific requirements for fatigue life, load magnitude and direction, dynamic response, and the angular range of motion across flexure and conventional bearing modes. To maximize the operational advantages of the wear-free flexure bearing, the mechanically programmed “transition point”, which acts as the boundary between the flexure and conventional bearings, should be selected such that the flexure bearing’s range contains the full range of motion over which high-cycle-count activities are expected. The flexure bearing and conventional bearing can then be independently selected or designed to achieve their respective angular ranges and support the maximum loads expected within that range. These bearings are then arranged in series, and the transition point between them is mechanically programmed via the retention mechanism.

Building from these principles, we designed a rotary MFB to support a primary angular range of 30 degrees and a secondary angular range of 270 degrees. We selected a cross-axis flexural pivot [23], [24], [25], [26], [27], [28], [29] bearing (model 5032-800, Riverhawk Co., New Hartford, NY, USA) with a range of ± 30 degrees, highlighted in teal in Fig. 2. To assess fatigue feasibility over this range, we used a reference stress of 625 MPa (the fatigue limit of AISI Stainless Steel 420 RT-DR for 2.0×10^4 cycles [30]); with a safety factor of 1.1, this gave us a final maximum allowable stress of 568 MPa. We intentionally chose a range that was within that final maximum allowable stress (see Supplementary Figure S2, which shows that the stress was 562 MPa at our allowed limits of ± 15 degrees, without any external loads). This unloaded stress comparison supports feasibility but does not establish high-cycle endurance. We selected a ball bearing (B180, igus® GmbH, Cologne, Germany) to serve as the conventional bearing and provide our target 270 degrees of secondary angular range, highlighted in yellow in Fig. 2. The transition point was programmed at 30 degrees of the rotary MFB’s overall angular range (we defined our angular measurements such that the equilibrium point of the flexure bearing was at 15 degrees).

This prototype rotary MFB was designed as a modular system capable of housing both the retention handoff and constant stiffness retention mechanisms (Fig. 2A). This had the added benefit of preserving inertial properties of the intermediate body across both retention mechanism experimental setups, such that dynamic performance could be compared directly between the two. We note for clarity that the flexure and conventional bearing positions are inverted in the constructed rotary MFB in Fig. 2 as compared to Fig. 1: in the constructed rotary MFB, the flexure bearing connects the ground body to the intermediate body while the conventional bearing connects the intermediate body to the stage. The only impact of this arrangement on device performance is on the inertia of the immediate body, which will change the threshold for entering a combined mode during dynamic operation, both arrangements are acceptable design choices.

Applying the design principle in Eq. 5 to the case of quasistatic operation, we considered the inertial torque and disturbance torque to be zero and designed the retention mechanisms such that the retention torque exceeded the restorative torque in the bearing at the end of its angular range, which was measured at 0.21 Nm. The retention handoff mechanism consisted of two sets of identical magnetic pairs (SB443-IN, K&J Magnetics, Inc., Pipersville, PA, USA), highlighted in magenta in Fig. 2, which were seated within dedicated notches in the rotary MFB housing. The magnetic pairs were at slightly different radii relative to the center of rotation, such that the outer pair provided a retention torque of 0.33 Nm and the inner pair of 0.25 Nm. The constant stiffness retention mechanism consisted of a torsion spring (Part No. 1842N5, McMaster-Carr, Elmhurst, IL, USA; magenta, Fig. 2) with a spring rate of 0.000180 Nm/deg, preloaded to 0.23 Nm of retention torque. Details on the assembly process and parts are provided in Supplementary Video S1 and Supplementary Fig S1. Videos demonstrating the manual manipulation of two rotary MFB configurations are available as Supplementary Video S2 (retention mechanism configuration) and Supplementary Video S3 (constant stiffness retention configuration).

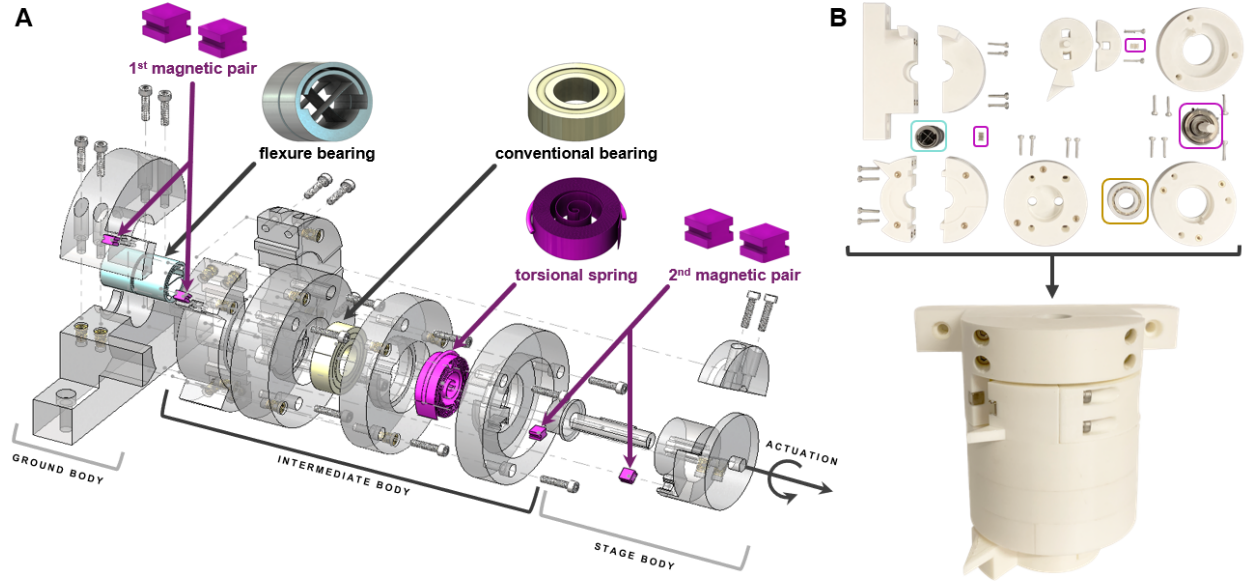


Fig. 2. Components of the example rotary MFB design. (A) Exploded model of the rotary MFB design featuring a flexure bearing (teal) and a conventional bearing (yellow) which tradeoff engagement aided by a retention mechanism (magenta). There are two retention mechanism options included in this design, a retention handoff (magnetic pairs) or a constant stiffness (torsion spring). Although both mechanism types are illustrated, only one is used at a time during testing. (B) 3D printed rotary MFB parts and assembly used for benchtop testing.

2.3. Experimental Design

We first characterized the torque-displacement behavior of the mechanism during quasi-static operation. During these experiments, the rotary MFB was mounted vertically to maintain consistent gravitational load across all displacements. The stage end of the rotary MFB was actuated at 5 deg/sec using a brushless motor (EC-I 52, Maxon, Sachseln, Switzerland) with a 10:1 planetary gearset (EG23-G10-D10). The ground body was fixed to the experimental frame (Fig. 3A). Real-time torque data was captured using a micro torque sensor (ATO-TQS-S04, ATO, Diamond Bar, USA). For video of benchtop data collection using the configuration in Fig. 3A, see Supplementary Video S4.

During operation, the distance between the hard stops of the intermediate body and the stage was measured using a magnetometer mounted adjacent to the stage-side magnet notch (Fig. 3A). This enabled us to monitor bearing mode in real time and to identify any combined-mode behaviors. This magnetometer tracked changes in the magnetic field at the stage-side hard stop, caused by proximity to the intermediate-side magnet (the intermediate-side magnet was left in place for the constant stiffness retention configuration for tracking purposes, even though it did not play a role in retention in that configuration). The magnetometer reading was highest when the stage body engaged the hard stop at the end of flexure-bearing range, positioning the intermediate-side magnet next to the magnetometer, and decreased as the distance between the hard stops increased. For each retention mechanism, the magnetic field readings were mapped empirically to angular span between the hard stops of the stage and intermediate bodies (See Supplemental materials document and Fig. S2).

Under dynamic operation, the torque required for retention increases due to the inertial loads from accelerating the intermediate body. Our second experiment evaluates the impact of increasing dynamic impulse on the operation of the rotary MFB with a fixed retention torque. According to the framework established in Eq. 1, in the absence of external disturbance torques ($\tau_d = 0$), only the internal inertial and flexure torques oppose the retention mechanism.

Therefore, for a given device architecture, the maximum tolerable angular acceleration can be characterized as an instantaneous breakout threshold whereby operation below this limit prevents the MFB from entering combined mode. The generalized dynamic predictive model for the breakout acceleration threshold is:

$$\ddot{\theta}_{threshold}(t) = \frac{\tau_r - \tau_f(\theta(t))}{J} \quad (6)$$

In the retention handoff mechanism configuration, the rotary MFB retention torque was $\tau_r = 0.33 \text{ Nm}$, the maximum flexure torque at 20 degrees (+5 degree deflection) was $\tau_f = 0.05 \text{ Nm}$, and the rotational inertia of the moving stage was $J = 9.676 \times 10^{-4} \text{ kg} \cdot \text{m}^2$. From Eq. 6, the predicted acceleration threshold for this specific configuration was 289.4 rad/s^2 (16,583 deg/s^2). In the constant stiffness configuration, the rotary MFB had a retention torque of 0.23 Nm, and identical flexural and inertial properties. From Eq. 6, the predicted acceleration threshold was 186.03 rad/s^2 (10,660 deg/s^2).

The aim of the dynamic portion of our experimental design was to validate the predictive model for the acceleration threshold of a given rotary MFB. Operating within flexure bearing mode, we used the motor to apply a controlled linear up-chirp of the bearing's angular position θ , resulting in increasing angular acceleration. The position chirp was generated from the following equation:

$$\theta(t) = -A \cos(2\pi(f_0 t + \frac{r}{2} t^2) + \phi_0). \quad (7)$$

The chirp rate was set to $r = 0.5 \text{ Hz/second}$, starting at $f_0 = 0 \text{ Hz}$ and terminating at $f_f = 10 \text{ Hz}$, over $t_f = 20 \text{ seconds}$. The amplitude A of the chirp was 5 degrees, and the initial phase ϕ_0 was 15 degrees, such that the system operated only within the flexure bearing's range throughout the entirety of the chirp (i.e. the conventional bearing should not have been active unless the mechanism entered a combined mode).

We can calculate the angular acceleration of the intermediate body during this chirp as

$$a(t) = \frac{d^2 \theta(t)}{dt^2} = A(2\pi r \sin(\Psi) + 4\pi^2(f_0 + rt)^2 \cos(\Psi)) \quad (8)$$

where:

$$\Psi = \pi t(2f_0 + rt) + \phi_0. \quad (9)$$

Because our goal is to calculate the maximum acceleration the prototype rotary MFB could withstand before entering combined mode (the "activation threshold"), it was necessary to calculate the peak acceleration experienced by the mechanism up to a given point in time. To start, we approximate that the peaks of the acceleration trajectory in Eq. 8 at the peak times (t_{peak}) will have the values:

$$a_{peak}(t) = 4\pi^2 A(f_0 + rt_{peak})^2. \quad (10)$$

Additionally, the number of completed cycles N of the chirp is approximately related to time t by the following equation:

$$N(t) = f_0 t + \frac{r}{2} t^2. \quad (11)$$

For any time t , the greatest acceleration experienced to that time will have occurred at the most recent acceleration peak t_{peak} , which corresponds to the nearest lower integer value of N (i.e. the floor function, $[N]$). For any value of N , the time at the most recent preceding peak t_{peak} is therefore (noting that ϕ_0 is small) approximately

$$t_{peak} = \frac{-f_0 + \sqrt{f_0^2 + 2r[N]}}{r}. \quad (12)$$

Plugging this into Eq. 10, we can derive an expression for the greatest acceleration experienced by the rotary MFB during the chirp up to any arbitrary time t :

$$a_{peak}[t] = 4\pi^2 A \left(f_0^2 + 2r[f_0 t + \frac{r}{2} t^2] \right). \quad (13)$$

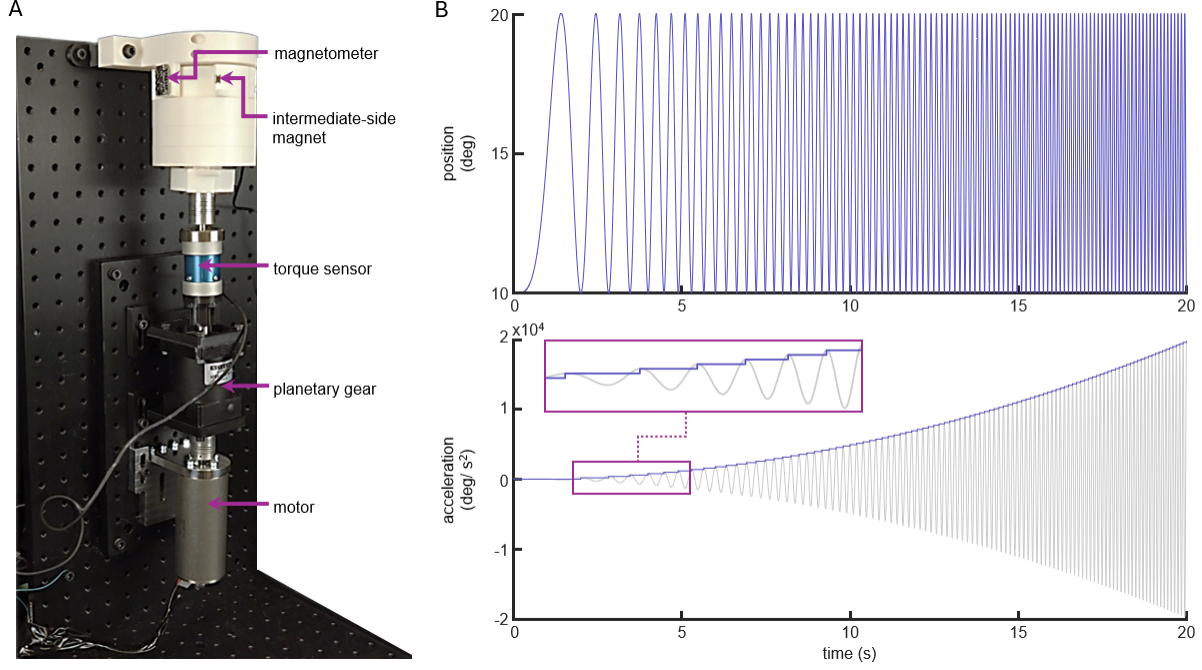


Fig. 3. Experimental setup of rotary MFB. (A) Benchtop setup for quasistatic and dynamic testing of both rotary MFB retention mechanism configurations: retention handoff and constant stiffness. (B) Angular position up-chirp and acceleration chirp (gray) with discretized acceleration threshold (purple inset). Note that we set our angular measurements such that the equilibrium point of the flexure bearing is at 15 degrees.

When the rotary MFB is operated in a pure flexure-bearing mode, the magnetometer readings, which correspond to motion of the intermediate body, can be expected to move in direct concert with the motion applied to the stage body (because the intermediate body is coupled to the ground body). Any deviation from this expected pattern can therefore be attributed to decoupled motion of the intermediate stage in a transient combined mode. In practice, we enforced a chirp pattern at the stage body and monitored the magnetometer reading for any meaningful deviation from the chirp signature. When such a deviation was detected, we then used Eq. 12 and Eq. 13 to calculate the greatest acceleration experienced by the rotary MFB to that point. This value was reported as the acceleration threshold for each configuration of our prototype rotary MFB.

Eq. 6 formulates the mode-change threshold as an instantaneous retention and breakout condition, rather than predicting sustained combined-mode motion. Redefining the dynamic equilibrium in Eq. 1 in terms of the breakout torque yields the relation:

$$\tau_{brk} = \tau_r + J\ddot{\theta}_{threshold} \quad (14)$$

Differentiating the breakout torque (τ_{brk}) with respect to the rotational inertia of the moving stage (J) yields a linear sensitivity:

$$\frac{\partial \tau_{brk}}{\partial J} = \ddot{\theta}_{threshold} \quad (15)$$

For retention mechanisms with inherent stiffness, the post-breakout transition angle (θ) behaves as a second-order spring-mass-damper system governed by system stiffness (k), internal damping (c), and stage inertia (J):

$$J\ddot{\theta} + c\dot{\theta} + k\theta = \tau_{brk}$$

Following instantaneous breakout, the transient response of the flexure bearing is governed by the damping ratio $\zeta = \frac{c}{2\sqrt{kJ}}$. Depending on the value of ζ , three distinct dynamic operational regimes emerge. The underdamped regime ($\zeta < 1$) will exhibit rapid release of the spring energy combined with low internal damping, causing transient overshoot and rebound across the retention threshold. Consequently, the bearing briefly crosses the instantaneous breakout boundary without settling, explaining why certain retention mechanism configurations may fail to enter a sustained combined mode. In the critically damped regime ($\zeta = 1$), energy imparted during breakout is dissipated smoothly without overshoot or dynamic rebound, facilitating reliable engagement into a sustained combined mode. In the over-damped regime ($\zeta > 1$), excess internal damping prevents dynamic rebound but introduces resistance which would delay mode engagement.

3. Results

3.1. Quasistatic Operation - Retention Handoff

The magnetic retention handoff mechanism exhibited three distinct behaviors across three “zones” of angular deformation. We designated the flexure-bearing zone, magnetic transition zone, and conventional-bearing zone based on the source of the observed behavior. When operating in the flexure-bearing zone, which spanned angular deformations from 0-20 degrees, the rotary MFB showed a near-linear stiffness consistent with that of the flexure bearing (Fig. 4, teal). At 20 degrees the rotary MFB entered a magnetic transition zone (20-40 degrees) during which the magnetic pull force altered the torque profile of the flexure bearing. The magnetic interactions between the two magnet pairs induced two peaks in the torque. The engagement of the first magnetic pair occurred at 26.6 degrees, observed as the negative torque peak (Fig. 4, point 1). Angular span distance was zero at this position, corroborating the engagement of the first magnetic pair (Fig. 4, bottom). Furthermore, this angular span remained at zero from 26.6-300 degrees confirming that the stage body and intermediate body remained continuously coupled over the remainder of the magnetic transition and conventional bearing zones. The second magnetic pair, which initially coupled the intermediate and ground bodies, disengaged at 30 degrees, which caused a positive peak in torque (Fig. 4, point 2). Note that the transition zone extends beyond the two torque peaks, due to ranged effects of the magnets as they drew together or apart. The nonlinear transition zone could also be explored by considering the energy in the system. Elastic energy stored in the flexure bearing was converted to kinetic energy during the magnetic handoff, generating the observed torque peaks and transient disturbance (Fig 4, bottom plot). The rotary MFB operated under isolated conventional-bearing mode from approximately 40-300 degrees, during which the rotary MFB exhibited zero stiffness.

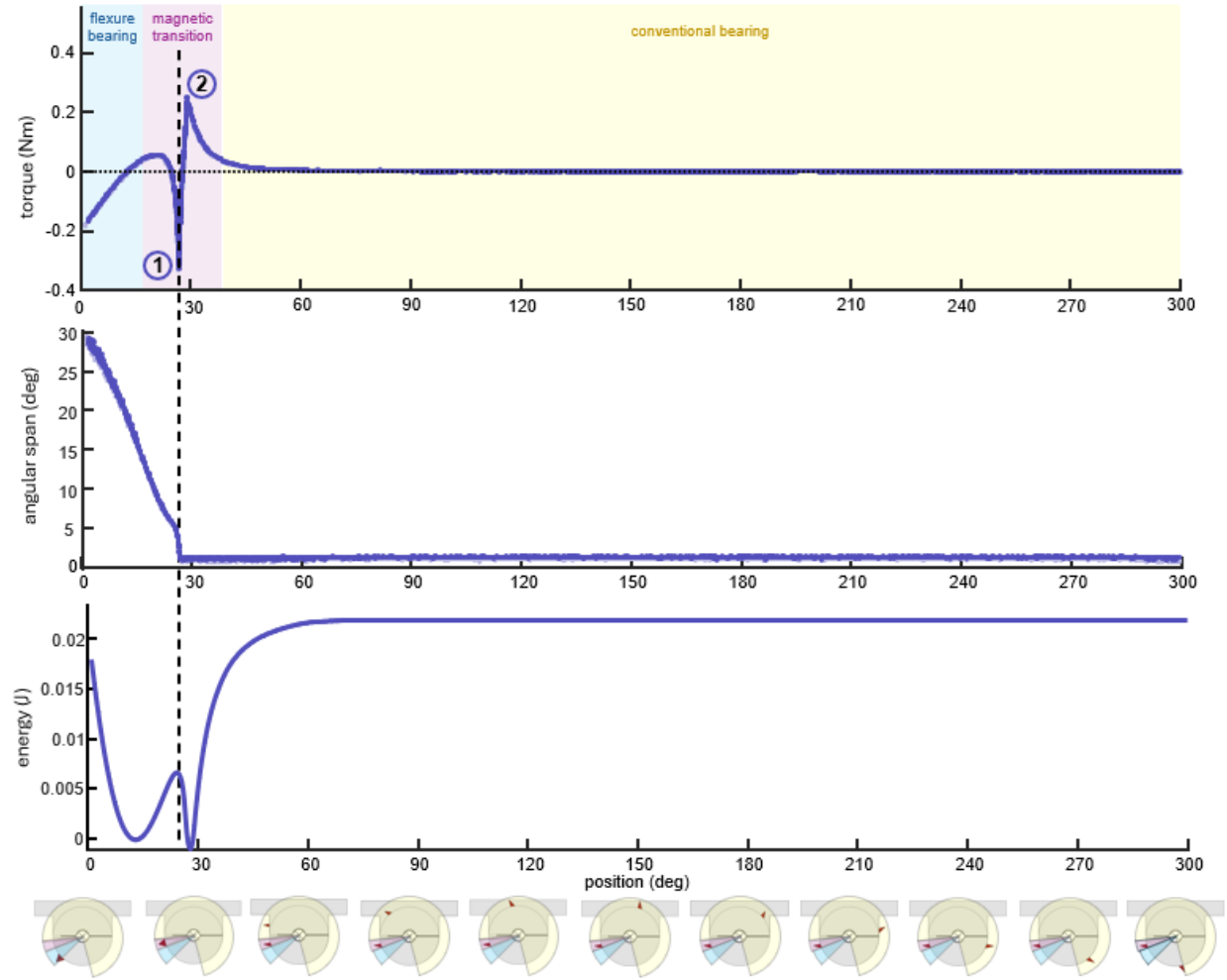


Fig. 4. Quasistatic behavior of retention handoff mechanism using magnetic pairs. Torque, angular span, and elastic energy are shown over the full rotary MFB range. Forward trajectory is plotted in dark purple and return trajectory is plotted in light purple. Point 1 shows interaction of the first magnet pair. Point 2 shows interaction of the second magnet pair. Vertical black dashed line represents the angular position of the transition. Horizontal black dotted line represents zero stiffness slope of conventional bearing. Illustrations of the rotary MFB correspond to the position axis and show body coupling and displacement with two red dials.

3.2. Quasistatic Operation - Constant Stiffness Retention

The constant stiffness retention mechanism exhibited two distinct zones of behavior: flexure-bearing zone and conventional-bearing zone. The rotary MFB operated under isolated flexure-bearing mode from 0-30 degrees, during which the rotary MFB exhibited constant stiffness consistent with that of the flexure bearing. The rotary MFB transitioned to the conventional-bearing zone at an instantaneous position, observed as a nonlinear increase of torque at 30 degrees (Fig. 5, dotted black line). The measured angular span distance at this position was zero, confirming hard stop engagement. As before, we see from the angular span measurement that the stage body and intermediate body remained continuously coupled throughout the full conventional bearing zone. The rotary MFB exhibits a constant non-zero stiffness throughout the conventional-bearing zone, consistent with that of the torsion spring. Considering the energy in the system, the presence of a torsion spring in the conventional bearing zone results in linearly increasing energy storage in the system during the conventional bearing zone (Fig 5, bottom plot).

As a result, the system requires increasing input torque to continue rotation, resulting in higher stored elastic energy that converts into greater kinetic energy during the return stroke.

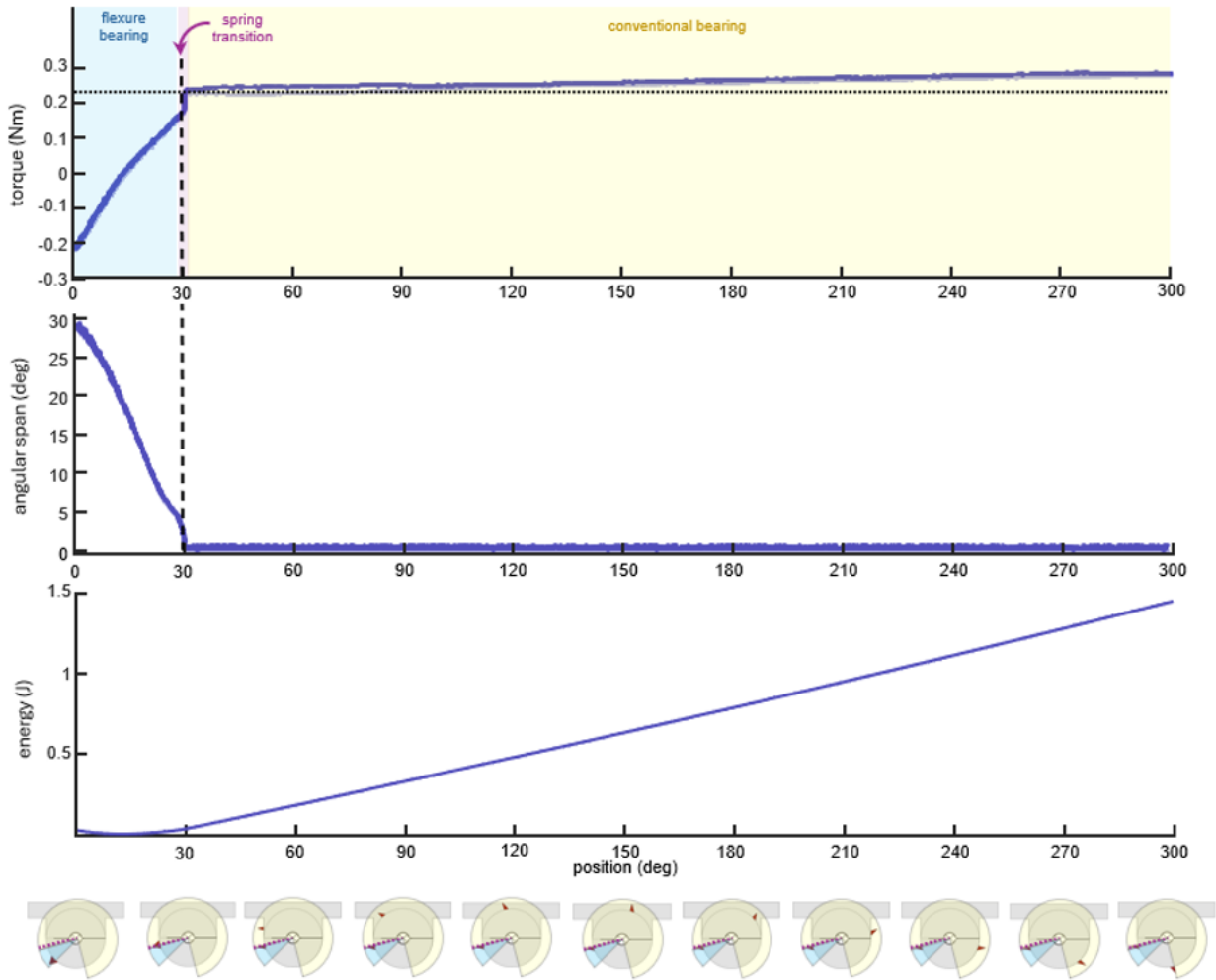


Fig. 5. Quasistatic behavior of constant stiffness retention mechanism using a torsion spring. Torque, angular span, and elastic energy are shown over the full rotary MFB range. Forward trajectory is plotted in dark purple and return trajectory is plotted in light purple. Vertical black dashed line represents the angular position of the transition. Horizontal black dotted line represents zero stiffness slope of conventional bearing. Illustrations of the rotary MFB correspond to the position axis and show body coupling and displacement with two red dials.

3.3. Dynamic Operation

In the magnetic retention-handoff configuration (Fig. 6A), the angular span initially exhibited a trend consistent with the position up-chirp pattern. Peaks in flexure bearing angle increased slightly with increasing frequency, demonstrating that greater impulses cause the device to move beyond the prescribed 10 degrees of rotation, but still within the flexure bearing's angular range. After 18.4 seconds, the angular span significantly diverged from the predicted chirp, indicating transient motion enabled by decoupling of the intermediate and ground bodies. This combined-mode behavior is highlighted in the green region in Fig. 6A, and an example device configuration is illustrated in the right column of Fig. 6B. The acceleration threshold of the rotary MFB retention handoff configuration was found to be $16,580.9 \text{ deg/s}^2$ (9.2 Hz) at 18.4 seconds. This acceleration threshold closely matches

with the predicted instantaneous threshold of $16,583 \text{ deg/s}^2$ found using our predictive model (Eq. 6). As frequency increased, the torque peaks and impulse also increased until exceeding the holding force of the flexure-side magnet and inducing combined mode. During combined mode, the device enters a state of chaotic motion, in which the components have unshared oscillations or resonance, and because we did not dampen our prototype system, it exhibited large torque peaks. We thus ended the experiment when it entered this mode to prevent the possibility of unwanted wear on the system. While the measured breakout acceleration aligned closely with our predicted value, this comparison was bounded by a single prototype benchmark and future multi-sample testing would quantify statistical repeatability and validate the sensitivity analysis models..

In the constant stiffness retention configuration (Fig. 6B), the gap distance followed a chirp pattern of linearly increasing peaks for all frequencies included in the sweep, indicating that the system never entered combined mode. While the position chirp reaches the theoretical acceleration threshold of $10,660 \text{ deg/s}^2$ at 14.7 seconds, the constant stiffness configuration does not transition observably into a sustained combined mode. Unlike the retention handoff mechanism, which utilizes a displacement-enforced disengagement, the constant stiffness mechanism relies on a continuous elastic restoring force. The condition of Eq. 6 established an instantaneous breakout boundary rather than predicting post-transition trajectory. Because our physical prototype exhibits low internal damping ($\zeta < 1$), the post-breakout dynamics are dominated by a rapid release of the elastic energy of the torsional spring, inducing transient overshoot and rebound across the retention boundary. The observed response of the constant-spring system is therefore consistent with the underdamped regime identified in our sensitivity analysis. The torsion spring notably introduced resonant behavior, seen in the torque plot as a reduction of torque around 9 seconds. The constant stiffness retention mechanism withstood a maximum impulse of $0.67 \text{ Nm}\cdot\text{s}$, which was greater than that of the retention handoff mechanism. Since the constant stiffness configuration never enters combined mode, no acceleration threshold was reported.

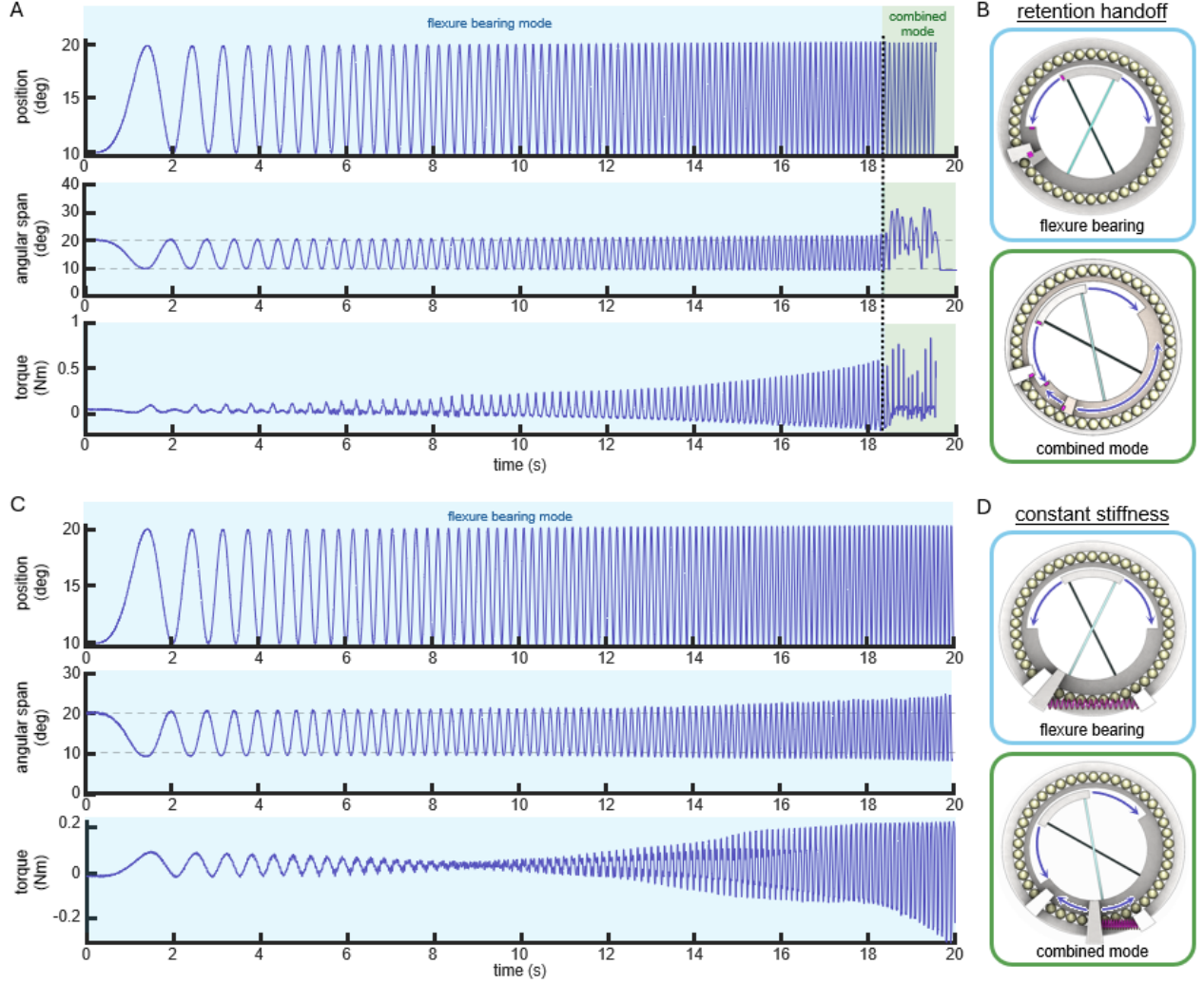


Fig. 6. Rotary MFB behavior during dynamic operation. Frequency sweep of position on the rotary MFB with resulting angular span (dashed lines indicate expected quasistatic angular span range) and torque for each of the retention mechanism types: (A) retention handoff and (B) constant stiffness retention. Illustrations of device configurations in flexure bearing and combined mode for (C) retention handoff and (D) constant stiffness retention.

4. Discussion

In this manuscript we introduce the rotary MFB, which is expected to provide high-precision, low wear compliant rotational motion over a limited range and transitions automatically to a larger-range conventional bearing when additional angular displacement is required. Our results show that the transition between bearing modes (flexure-bearing to conventional-bearing and vice versa) happens predictably and reliably as a function of the retention mechanism's design parameters. This behavior was demonstrated across two different retention mechanisms, with distinct force-displacement profiles in the transition zone and conventional-bearing mode. We also demonstrated the emergence of a "combined mode" under dynamic operation, and established principles for determining when this combined mode would emerge. In the following sections, we describe a systematic framework for design and optimization of the rotary MFB, introduce several design variations that impact the rotary MFB's functional properties, and discuss key considerations in selecting a retention mechanism for a specific application.

4.1. Designing a Rotary MFB for a Specific Application

The first step in our design framework involves defining key performance specifications such as primary (flexure-bearing) range of motion, secondary (conventional-bearing) extended range of motion, expected load magnitude and direction during typical operation in each bearing mode, angular displacement from the undeformed flexure bearing at which transition occurs, and any requirements for transitional behavior (e.g. smoothness). If the bearing will be part of a larger system, it may also be important to define specifications for volume, mass, and other packaging considerations. Having established these requirements, we first recommend selecting or designing a compliant mechanism in isolation that can meet the load/deformation requirements of the flexure bearing stage with peak stresses below the material fatigue limit. For applications with high mechanical demands, we recommend following a stress-based optimization protocol similar to the one described in [16], though there are many available design strategies for compliant mechanism design, including in particular strategies for designing cross-axis flexural pivots [31], [32], [33], [34], [35], [36], [37], [38], [39], [40]. Recent developments include the TRIVOT for reduced parasitic center shift [41], curved cross-axis pivots for improved accuracy [42], [43], and triple-axis pivots that balance axis drift, rotational stiffness, and stress [44]. Other optimized architectures include corrugated cross-axis pivots [45] and pivots combining near-zero center shift with large motion range and high bearing stiffness [46]. These developments provide additional options when selecting or designing the rotary MFB's flexure stage. In the present study, we selected an off-the-shelf flexure bearing with a range of ± 15 degrees and operated under zero compressive or tensile load. We did not measure system-level precision, backlash, or lifetime; claims about these properties refer to expected performance of a rotary MFB with a well-designed flexure bearing. Once the flexure bearing has been selected, we recommend a similar protocol for identifying or designing a conventional bearing capable of supporting the loads and displacement required across the extended range of motion. In this study, we selected a ball bearing that was capable of free motion to full 300 degrees of displacement in our established criteria. A key takeaway in this framework is that the two bearing stages can be designed independently of each other because they operate independently, so long as the bearing does not enter combined mode. We recommend starting with the flexure bearing because in our experience this puts greater strain on the volumetric requirements than the conventional bearing, which can typically be smaller.

With the two bearings in place, the next consideration is the design of the retention mechanism. The most important insight in retention mechanism design is that the retention torque must exceed the maximum restorative torque produced by the flexure bearing at the end of its range of motion. Positioning of the retention mechanism across the rotary MFB's range of motion determines the onset of transition between the two mechanisms. The choice of retention mechanism type has a notable impact on two important elements of the rotary MFB's mechanics. First, different mechanisms induce different transitional behaviors, which may affect the angular displacement at which transition occurs. For instance, although our flexure bearing was capable of a full 30 degree range of motion, our results show that with the magnetic retention handoff mechanism, the rotary MFB only operated under flexure bearing mode up to 20 degrees, after which the magnetic forces created a transition zone from 20.0-26.6 degrees, characterized by highly non-linear force-displacement behavior. The constant-stiffness retention mechanism has no such transition zone, and transition happened precisely at 30 degrees. Second, different mechanisms are associated with different behaviors in the conventional bearing mode. The magnetic retention handoff mechanism applied no sustained torques in conventional-bearing mode, allowing for a theoretically infinite extended range of motion. The constant-stiffness retention mechanism, on the other hand, is characterized by a sustained restorative torque after transition to the conventional bearing, and the deformation of the constant-stiffness spring will be bound by material limits.

Another key distinction between the retention-handoff and constant-stiffness retention mechanisms is the sustained presence of a resistive torque throughout conventional bearing mode. The constant-stiffness retention mechanism introduces a resistive torque at least as large as the retention torque throughout the entire conventional bearing mode. In contrast, the retention-handoff mechanism carries no sustained resistive torque, thereby reducing the actuation effort required during operation in conventional-bearing mode. However, our dynamic analysis revealed that the absence of sustained resistance torque makes the system more susceptible to decoupling and entering a

combined mode during dynamic operation. As such, the retention-handoff mechanism requires a greater retention torque to achieve the same acceleration threshold, which introduces a more pronounced non-linear transition zone. For these reasons, it is important to consider the operational requirements (dynamics, smoothness of transition, presence of sustained resistance torque, etc.) of the specific application when choosing a retention mechanism; this is the focus of Section 4.2.

Once the retention mechanism is established, the stage, intermediate, and ground bodies can be designed. As with any bearing system, these should be designed as a tradeoff between strength and mass, volume, or material cost, depending on the application. With these frame elements finalized, the acceleration threshold of the device can then be determined through either dynamic testing or analytical calculation of the impulse required to overcome retention torques. This acceleration threshold can be increased by increasing the retention torque beyond the minimum required to oppose the restorative torque in the deformed flexure bearing (e.g. stronger magnets or stiffer springs). However, any increase in the retention torque will create a step in the force-displacement curve at the transition point, which may cause a non-trivial “stickiness” during operation. In sensitive dynamic applications, this stickiness may be a necessary consequence of needing to avoid combined mode, in which the intermediate body is neither grounded nor actuated. Combined mode may be problematic because it decouples the location of the intermediate body from that of the stage body, such that the proportion of displacement in the flexure bearing versus the conventional bearing is not controlled. This can also cause the conventional bearing to be engaged during cycles that would typically only require flexure bearing mode, creating unnecessary wear. However, in applications that are less sensitive to wear, strategically entering combined mode could also allow the flexure bearing to remain in its undeformed state (not bent) in the presence of occasional high loads, which could reduce stress in the compliant mechanism and increase longevity.

Our prototype rotary MFB was designed without considering axial or transverse loads, ensuring that torque output was only a product of the retention mechanism and that only inertial effects were measured for acceleration threshold. In applications where external loads are present, additional emphasis would be placed on bearing load ratings. Over the entire operation of the rotary MFB, the flexure bearing will be subject to external load. In cases of uniform load, the flexure bearing will experience maximum load at its greatest deformation. For cases of non-uniform loads, the ranges of each bearing mode can be altered to optimize for bearing strength. Flexure bearings are stronger when there is no blade deformation, thus the mode range would skew such that maximum loads occur at a more neutral blade deflection. Further design optimization would be required for these applications, but the principal behaviors of the retention mechanism and primary bearing modes described in this article are still pertinent.

For applications where bidirectional range extension is needed, it is possible to design a rotary metamorphic flexure bearing for extended range in both directions. Strategies for doing so are discussed in our previous work [21].

4.2. Selecting a Retention Mechanism

The rotary MFB has the potential to improve machine performance in applications that would benefit from the accuracy, precision, and longevity of compliant mechanisms, but occasionally require larger displacements than are possible with a flexure. For instance, a pick-and-place machine could utilize the flexure bearing mode for highly repetitive small displacements, such as high precision pick-and-place tasks, and then switch to conventional bearing mode for less-frequent, large-displacement repositioning motions. A rotary MFB could be incorporated into a steering wheel for which typical turning motion is achieved under flexure bearing mode, but large turns engage the conventional bearing. This architecture could also be designed as a force-limiting clutch for a robotic end effector and used to enhance human safety during human-robot interaction. In most applications, the rotary MFB will function as part of a larger machine, introducing additional design considerations such as volume and mass, transition behavior, precision, and effort required for actuation. Here, we explore these considerations in the context of the two retention mechanisms we tested in our study, based primarily on our experimental results.

Many applications, such as those in aerospace, biomedical devices, or micro-electromechanical systems, are tightly constrained in volume and/or mass, such that these become important considerations in the rotary MFB design. In general, implementation of the magnetic retention handoff mechanisms tends to require less mass and volume than the constant-stiffness mechanism, simply because the maximum energy storage in the retention spring (which dictates retention force and range of motion) increases with spring mass and volume. Additionally, we found that the magnets were easier to package in the overall system, whereas the retention springs introduced greater complexity into the frame design. In our prototype rotary MFB, the volume of the torsion spring (0.8835in^3) was 1800% larger than the total volume of the four magnets (0.047in^3), without any consideration of the added packaging complexity. Similarly, the mass of the torsion spring (1.65oz) was 780% greater than the combined mass of the four magnets (0.1868oz). The reduced form factor achievable with the magnetic retention handoff mechanism may make it better suited to applications involving limited mass or volume than the constant-stiffness retention mechanism.

Some applications, such as those involving physical interaction with humans, might place high emphasis on transition point behavior, because abrupt changes in torque-displacement behavior may be perceivable. In such cases, a smooth transition between bearing modes may be of a high enough priority to drive selection of a retention mechanism. In general, the constant-stiffness retention mechanism will tend to feel “smoother” than the retention handoff mechanism, because the former does not create a non-linear transition zone. For an extended exploration of the topic of transition regions, see our previous work [21]. In our prototype rotary MFB, the constant stiffness retention mechanism featured a smaller instantaneous torque disturbance of approximately 0.05 Nm, whereas the retention handoff generated two torque disturbances with peaks of 0.4 Nm and 0.6 Nm. The constant-stiffness retention mechanism’s transition can be made nearly imperceivable by matching bearing and retention mechanism stiffnesses and calibrated preloading, resulting in a continuous linear torque profile the entire range of motion.

Other applications that require a smooth transition may be sensitive to the presence of a sustained retention torque. In this case, retention handoff can be used, but the retention torque should be optimized to exactly match the torque required to maintain coupling of the intermediate body to the stage or the ground bodies. This minimizes (or ideally eliminates) the torque disturbance but makes the rotary MFB more likely to enter a combined mode during dynamic operation. Retention handoff mechanisms other than the magnets we used may be better suited to reducing torque disturbance. For example, a bistable mechanical latch could achieve body coupling with a much smaller transition zone than the magnets because there would not be magnetic attractive forces prior to the transition point.

In yet other applications, especially those involving physical interaction with a human, the presence of a torque disturbance at the transition point may be desirable. Such a perceivable shift in mode could serve as haptic feedback to indicate that the mechanism has been displaced beyond a certain angular range. In this case, a retention handoff mechanism might be most appropriate. The mode transition could also be used as a mechanical fuse for operational safety. In such applications, it may be helpful for the rotary MFB to produce a perceivable transition out of flexure bearing mode and into a conventional bearing mode with zero stiffness. This could be used for instance as a safety feature on an end-effector that does not allow joint torque to exceed the retention torque. To produce this behavior, a constant-force retention mechanism with a large retention torque offset (retention torque much greater than restorative torque in the flexure bearing at maximum deformation) would be most appropriate.

In this manuscript, we introduced only two of the many possible types of retention mechanisms. In fact, any mechanism capable of imposing a controlled force on the intermediate body to couple it to either the ground or the stage bodies can serve as the retention mechanism. Other options for retention handoff beyond those explored in this work include mechanical or inertial catches, actuated release mechanisms, friction plates or collars, or spring-loaded detents. Other constant-stiffness retention options include a variable radius cam to a suspended mass, a gas spring, or a powered actuator programmed to act as a spring. There also exist other categories of retention mechanisms beyond those that were explored in this manuscript, such as constant-force and non-linear stiffness retention mechanisms. These variations would produce sustained torques in the conventional bearing mode, but the torque-displacement profiles could be tuned to match desired behavior.

5. Conclusion

The rotary MFB demonstrates a new paradigm for rotational bearings that integrate compliant and conventional bearing components within a single architecture. This approach is expected to enable low-wear, high-precision operation during high-cycle-count tasks while preserving the capacity for large angular excursions as needed, thereby addressing a fundamental tradeoff in range of motion that is inherent to flexural bearings. The framework and design principles introduced here provide a foundation for tailoring rotary MFB behavior across applications, and future work will focus on long-term durability, packing, and optimization for specific applications. Together, these results position the rotary MFB as a promising next-generation bearing technology across a broad range of engineered systems.

CrediT authorship contribution statement

Alyssa Tomkinson: Writing – original draft, Visualization, Validation, Formal analysis, Data curation, Conceptualization, Methodology, Investigation. **Cameron R. Taylor:** Writing – review and editing, Conceptualization, Validation, Methodology, Investigation. **Will Flanagan:** Writing – review and editing, Software, Resources, Data curation. **Jonathan B. Hopkins:** Writing – review and editing, Conceptualization, Visualization. **Tyler R. Clites:** Writing – review and editing, Conceptualization, Visualization, Project Administration, Supervision, Funding acquisition

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Displacement-enforced mode transitions in rotary metamorphic flexure bearings

Supplementary Materials

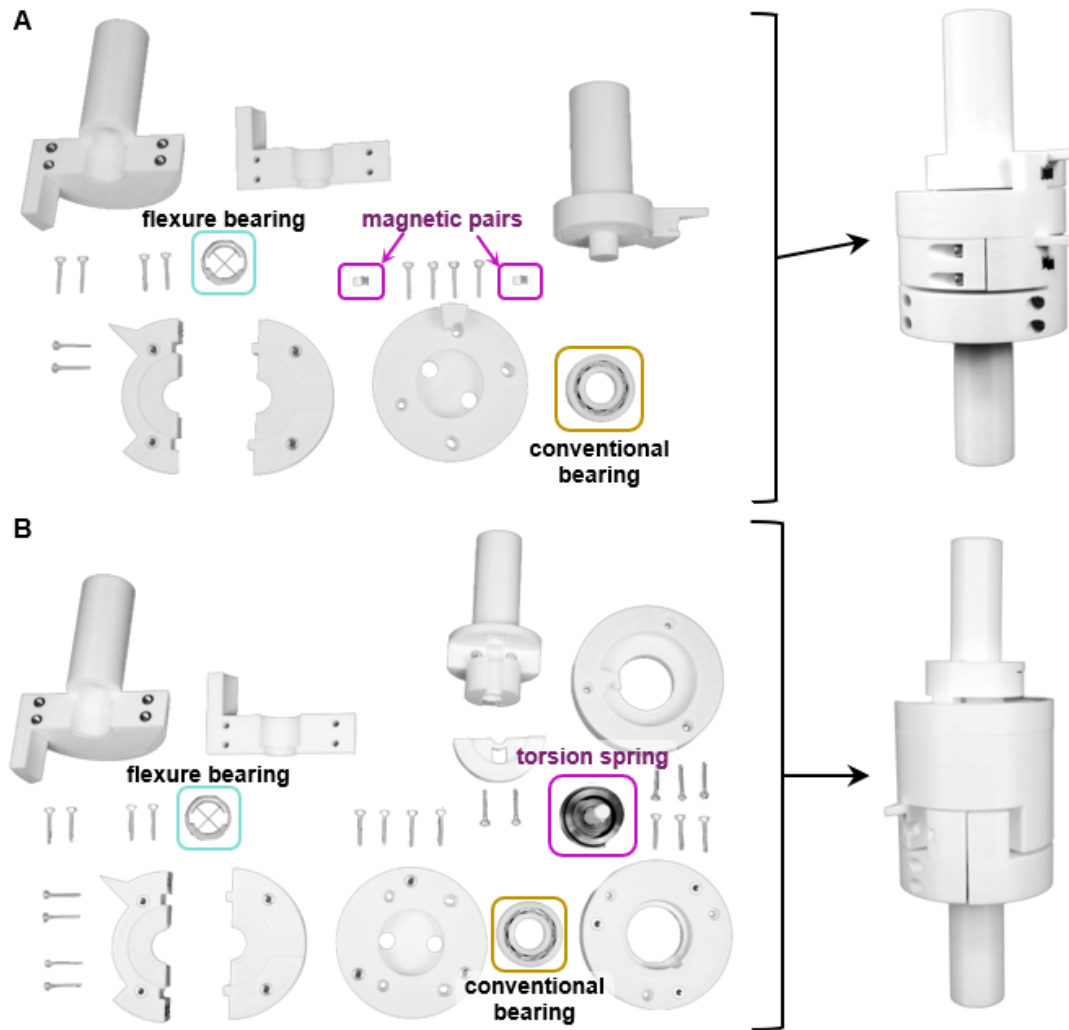


Fig. S1. All components of a 3-D printed RMFB laid out and assembled. These models were designed with the same intermediate body and inner components of the benchtop testing mechanism; however, the stage body and ground body were modified with handles such that we could rotate the mechanisms manually and explore “feel” of the bearing performance.

Flexure Bearing Optimization

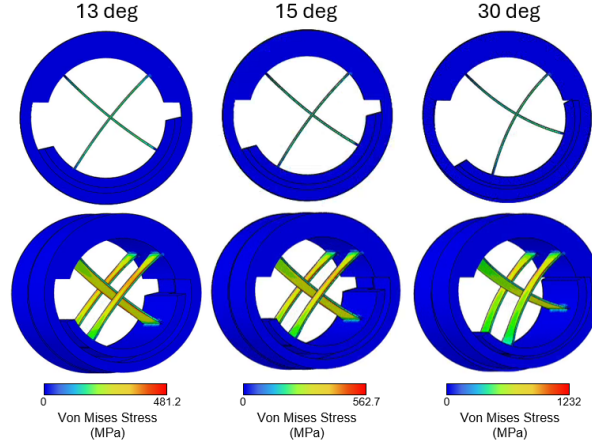


Fig. S2. Finite element analysis performed using Abaqus (Dassault Systèmes, Providence, RI, USA) demonstrating peak stress experienced in the flexure pivot for 13, 15, and 30 degree deflections.

Empirical Linearization of Magnetic Flux

To accurately correlate magnetic sensor data with the angular span between the ground and intermediate bodies, an empirical linearization process was implemented to overcome non-linearities inherent in near-field magnetic sensing and rotary kinematics. Hall effect sensors follow an inverse-cube law of magnetic field decay as the magnet distance increases,

$$B \propto \frac{1}{d^3}. \quad (1)$$

resulting in non-linear magnetic flux readings (seen in Fig. S3.A). Additionally, since the ground and intermediate stages are concentric and move angularly relative to each other, there are additional non-linear effects from this rotary kinematics. The raw data was pre-processed by calculating the Euclidean norm of the three-axis magnetometer readings

$$B_{Total} = \sqrt{B_x^2 + B_y^2 + B_z^2}. \quad (2)$$

This ensures the measurement is invariant to the rotation of the magnetic field vector as the magnet travels along the circular arc of the bearing. The data was isolated to the "active region", defined by the motor positions corresponding to the initial 30 degree offset through to the mechanical hard stop. Because theoretical inverse-cube models often fail to account for sensor saturation and complex magnet geometries in proximity, a third-order polynomial fit was utilized. This polynomial maps the non-linear flux magnitude directly to a target linear profile representing the known displacement sweep. The full set of magnetometer data was then mapped using the polynomial (Fig. S3B).

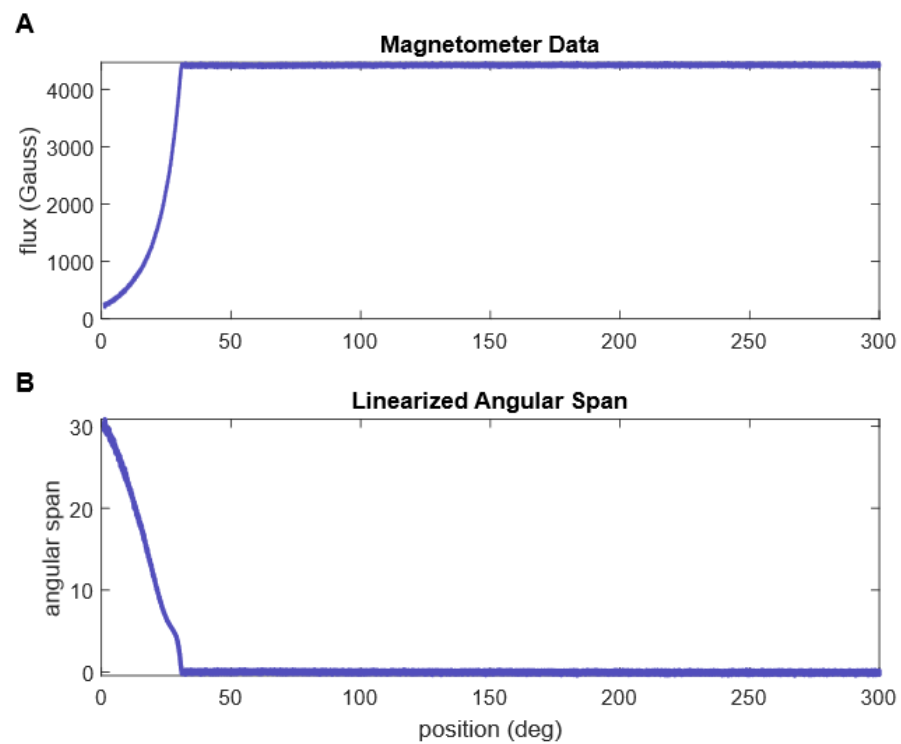


Fig. S3. (A) Magnetometer readings in Gauss. (B) Linearized angular span data derived from the magnetometer readings